

Due Date: 11:59 pm on 2/7/2026

Instructor: Dr. Tutku Karacolak

ECE 370 Electromagnetic Fields and Waves

HOMEWORK #2

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Problem	Course outcome	Grade
1	1-a, 1-d	/40
2	1-a, 1-c	/30
3	1-a, 1-d	/30
TOTAL:		/100

- 1- A 7 C charge is surrounded by a thin spherical shell with a radius of a . The shell carries a charge density of 3 C/m^2 . The shell is further surrounded by a dielectric material (inner radius b , outer radius c) with a volume charge density of $5R \text{ C/m}^3$ and dielectric constant ϵ_r . The structure is shown below (Figure 1). Find the electric field intensity in the following regions:

- (a) $R < a$
- (b) $a < R < b$
- (c) $b < R < c$
- (d) $R > c$

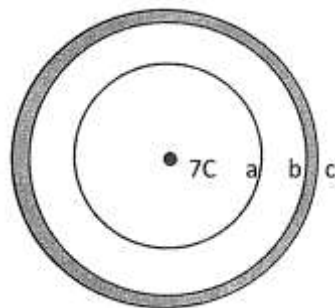


Figure 1.

- 2- Charge $q_1 = 6 \mu\text{C}$ is located at $(1 \text{ cm}, 1 \text{ cm}, 0)$ and charge q_2 is located at $(0, 0, 4 \text{ cm})$. What should q_2 be so that electric field ($\vec{E}$) at $(0, 2 \text{ cm}, 0)$ has no y component?

- 3- Charge Q_1 is uniformly distributed over a thin spherical shell of radius a , and charge Q_2 is uniformly distributed over a second spherical shell of radius b , with $b > a$. Apply Gauss's law to find $\vec{E}$ in the regions $R < a$, $a < R < b$, and $R > b$.